				[2]			
12	**(i)**		$A = 500$ $k = 1.044$	**B1** **B1** **[2]**	**3.3** **1.1**		
12	**(ii)**		500×1.044^{10} = 769.1 which is not close to 650 so not consistent	**M1** **A1** **[2]**	**3.4** **1.1**	Ft www	May use 650 and show k or A different
12	**(iii)**		exponential growth oe	**B1** **[1]**	**3.5a**	OR increase for ever, oe	

Question			Answer	Marks	AOs		Guidance
12	**(iv)**		$t = 0$, $N = 500$ $t = 10$, $N = 650.37 \approx 650$ model predicts number of fruit flies tends to 750 in the long run	**B1** **B1** **B1** **[3]**	**1.1** **1.1** **3.5a**	Oe 'N will not go beyond 750'	Allow shown using large value of t
12	**(v)**	(*A*)	Oscillations	**B1** **[1]**	**3.5b**	oe	
		(*B*)	cosine or sine oe	**B1** **[1]**	**3.5c**	Eg Introducing a multiplier of $(-1)^t$ if t is a large integer.	